

Modeling and Monitoring of Within-Match Fatigue in Professional Football

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Background

Managing player fatigue in real time, in particular when making substitution decisions, is a key challenge for coaching staff in professional football.

The Data

Minute-by-minute GPS and accelerometer data across 35 matches for 28 Hibernian FC players during the 2025/26 Scottish Premiership season.

- Total Distance Covered (m)
- High Speed Distance Covered (m), exceeding 19.8 km/h
- Max Velocity (m/s²)
- High-Intensity Accelerations (count), velocity exceeding $\pm 3\text{m/s}^2$
- High-Intensity Decelerations (count), velocity exceeding $\pm 3\text{m/s}^2$

Data restricted to starting players.

Fatigue Operationalization

Fatigue as performance degradation is defined as the reduction in explosive outputs and the failure to maintain required/expected output [1].

High intensity accelerations and decelerations are established indicators of within-match physical fatigue [2] and showed the clearest match related decline of all available GPS metrics and was thus chosen as the basis for the fatigue measure.

Explosive Output Score

10-minute rolling sum of explosive actions:

$$F_{i,t} = \sum_{k=t-9}^t (\text{HIA}_k + \text{HID}_k)$$

First-Half Peak

90th percentile of explosive output score across first-half minutes:

$$\text{Peak}_i = Q_{90}(F_{i,t} : t \in \text{H1})$$

Deterioration

Proportional decline below first-half peak at minute t :

$$D_{i,t} = \frac{\text{Peak}_i - F_{i,t}}{\text{Peak}_i + \varepsilon}$$

i = player, t = match minute

Validation of Deterioration Metric

75.2% of players substituted out had declined to below 50% of their first half-peak, compared to only 30.4% of players who completed the full match.

Players removed from the pitch showed substantially higher deterioration at the point of substitution than players who completed the full match.

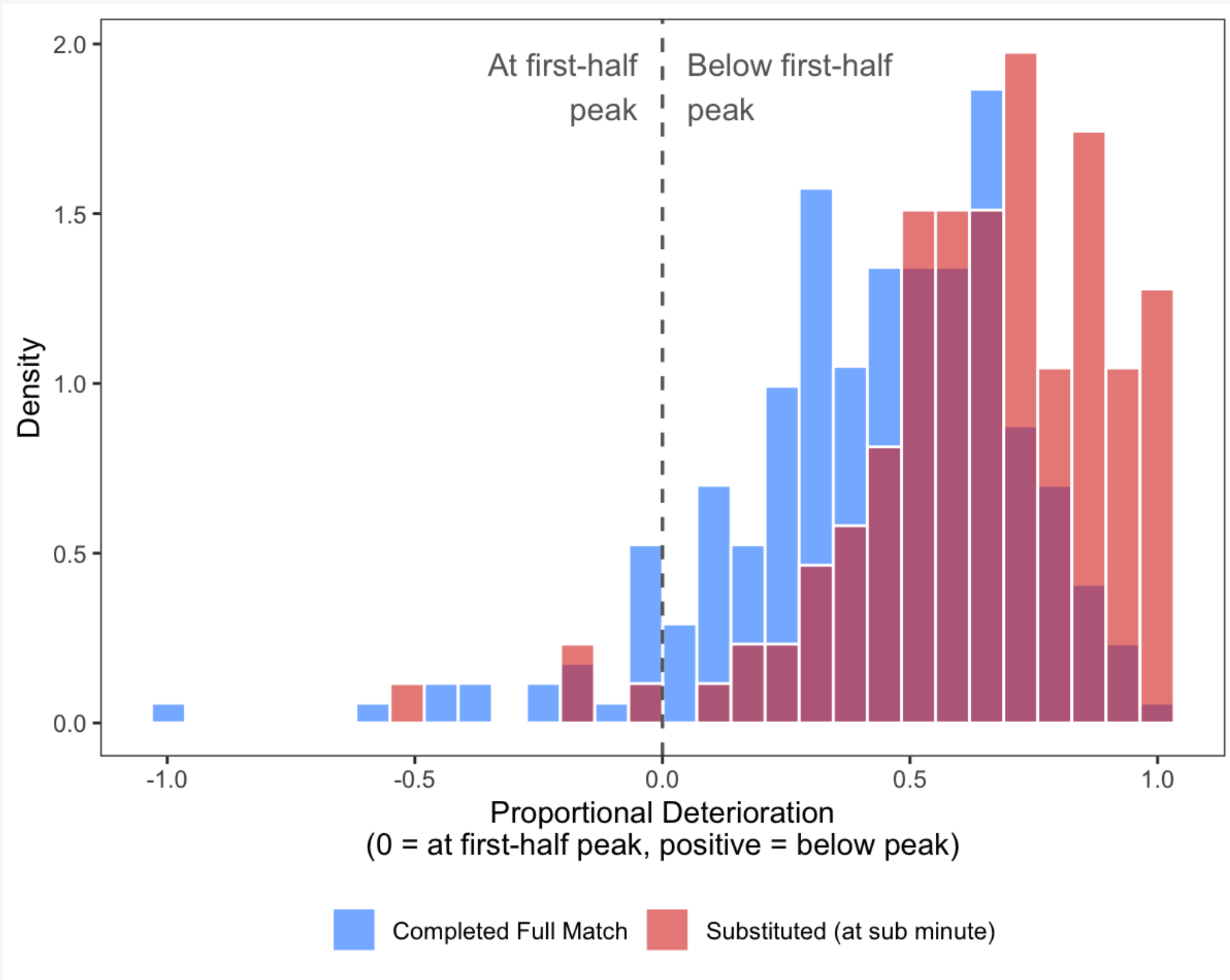


Figure 1. Distribution of deterioration at the point of substitution for players removed during the second half vs deterioration at the final recorded minute for players who completed the full match.

Modeling Framework

Data was split into 28 training matches and 7 test matches for performance evaluation to ensure results reflect predictive ability.

Models were fit to predict each players deterioration at 1, 5, and 10 minutes ahead by adding corresponding lag terms.

Five model specifications tested: Linear, Quadratic, Cubic, and Generalized Additive Model (Gaussian and Scaled t).

Predictors available to each model:

- Current fatigue level (lagged deterioration)
- Match minute
- Days Since Last Match
- Days into Season
- First Half Peak
- Cumulative Total Distance
- Cumulative High Speed Distance
- Players position
- Venue (Home/Away)

Player and match random effects included to account for individual differences between players and matches.

Predictive Results

Models were compared on their predictive ability. The linear model consistently matched or outperformed more complex alternatives on unseen test matches across all three forecast lengths and was thus selected on grounds of parsimony and performance.

Linear Model Performance

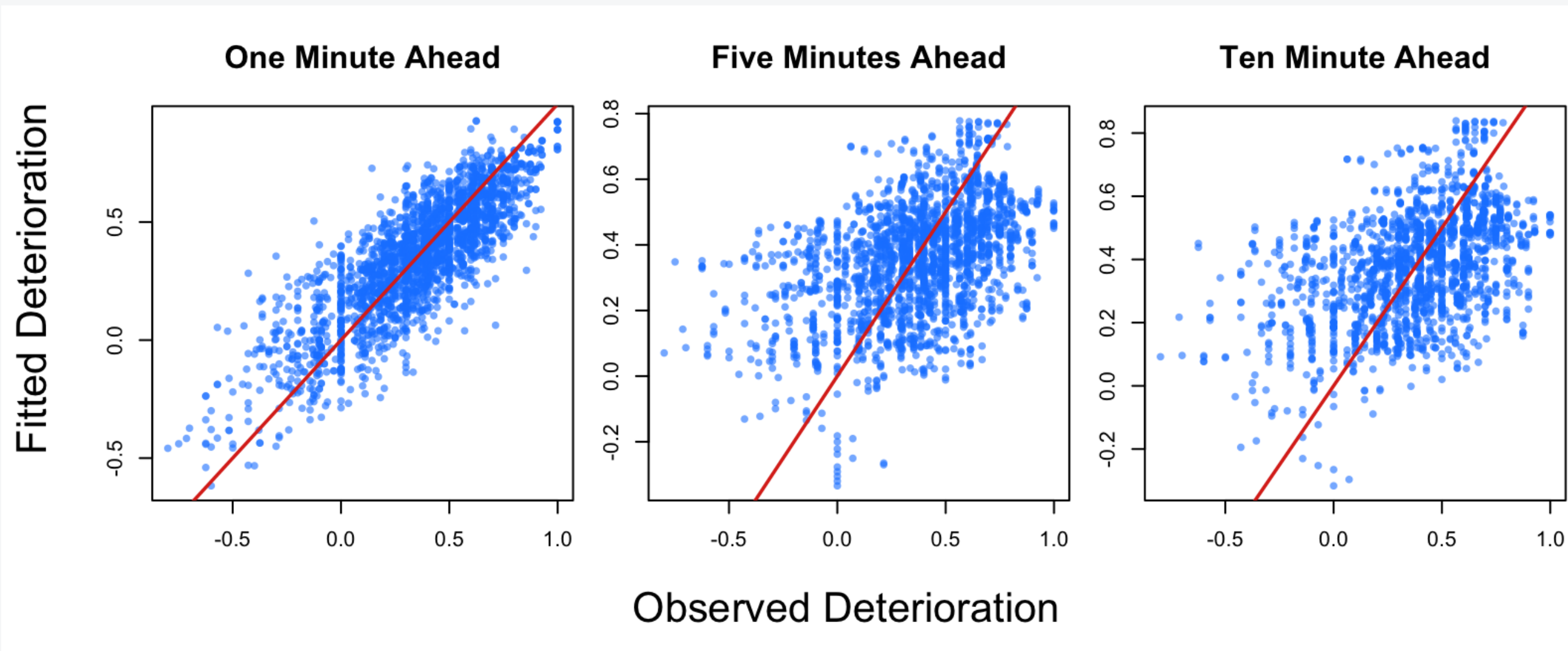


Figure 2. Observed versus fitted deterioration for the selected linear mixed effects model at forecasts of one, five, and ten minutes ahead.

Accurate prediction of future deterioration becomes increasingly difficult at longer forecast lengths as the precision declines notably beyond one minute ahead.

Flagging Accuracy

For **89%** of minutes, at least 3 of 4 most fatigued players correctly identified for *1 min ahead* predictions. **57%** and **65%** at *five* and *ten* minutes ahead respectively.

Applications to Real Time Insights

Provide coaching staff with an objective and interpretable view of player fatigue states, identifying which players to keep an eye on, and who may warrant prioritization for substitution.

Hibernian FC Half Time Fatigue Report Match 4006418 Score 1-2 (Losing)

Rank	Player	Position	Deterioration
1	Warren O'Hora	CB	55.5%
2	Dylan Levitt	DM	50.0%
3	Jamie McGrath	AM	50.0%
4	Nicky Cadden	FB	40.7%
5	Jack Iredale	CB	25.0%

Figure 3. Ranking of top 5 players most deteriorated at end of first half to inform second half planning.

Hibernian FC Fatigue Monitor Minute 68 Match 4006500 Score 2-1 (Winning)

Rank	Player	Position	Deterioration
1	Grant Campbell Hanle	CB	61.5%
2	Josh Mulligan	FB	61.5%
3	Jordan Obita	CB	40.0%
4	Daniel Barlasar	DM	30.7%
5	Martin Boyle	CF	18.2%

Figure 4. Live second half fatigue ranking of top 5 players most deteriorated at that minute.

Conclusion

- Substituted players showed significantly higher deterioration than those who completed the full match, validating deterioration measure.
- Precise predictions of future deterioration is beyond the capability of GPS metrics alone, models instead identify which players are most at risk.
- Framework provides objective physical evidence to support coaching judgment, not replace them.

Limitations & Future Work

- External GPS Metrics capture mechanical demands only. Internal load data such as heart rate could provide a more complete picture of physiological fatigue.
- Single season dataset (35 matches) limits generalizations. Results should be interpreted with caution given small test set of 7 matches.

Future Research: Integrate internal load metrics and expand to multiple seasons.

References

- R. H. T. Edwards, "Biochemical bases of fatigue in exercise performance: Catastrophe theory of muscular fatigue," *Biochemistry of Exercise*, vol. 13, no. 13, pp. 3–28, 1983.
- D. J. Harper, C. Carling, and J. Kiely, "High-intensity acceleration and deceleration demands in elite team sports competitive match play: A systematic review and meta-analysis of observational studies," *Sports Medicine*, vol. 49, no. 12, pp. 1923–1947, 2019.